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Constellations of Solidarity: Mutual Aid's Impact in Humanitarian Crises Caused by Conflict vs. Disaster

Analysis of Locally Led Responses in Sudan and Spain (2023–2025)

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Abstract

How does the impact of mutual aid in humanitarian crises differ in contexts of armed conflict vs. natural disaster? This thesis argues that grassroots efforts are influenced by the type of crisis in which they develop, taking Sudan's Emergency Response Rooms (2023-2025), and volunteer networks created after the DANA floods in Valencia, Spain (2024) as the main case studies. Additional insights from Ukraine's war mutual aid initiatives (2022-2025) and volunteer responses to Al-Haouz's earthquake in Morocco (2023) are provided to increase the validity of results. The research draws on Kropotkin's Mutual Aid Theory, interpreted from a sociological lens, to show how locally led responses can build civil society's resilience. Using a qualitative methodology, a comparative analysis design, and data collected from interviews and literature, this study investigates the benefits, limitations, and long-term effects of mutual aid in both settings. Findings reveal that natural disaster scenarios show more advantages and favorable potential, whereas conflicts present greater challenges and constraints. The long-term impact of mutual aid can lead to community empowerment, but conclusions from ongoing crises remain speculative. Lastly, recommendations are provided to advocate for greater integration of mutual aid approaches in humanitarian planning and policy.

Key words: *mutual aid, armed conflict, natural disaster, Sudan, ERR, Spain, DANA volunteer networks, locally led responses, civil society resilience.*

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List of Abbreviations

ERR	Emergency Response Rooms
NGO	Non-Governmental Organization
MDSD	Most-Different System Design
TA	Thematic Analysis
CSO	Civil Society Organization
RSF	Rapid Support Forces
SAF	Sudanese Armed Forces
OCHA	United Nations Office for the Coordination of Humanitarian Affairs

1. Introduction

This thesis explores the impact of mutual aid in humanitarian crises, focusing on a comparative approach between grassroots responses to armed conflict and natural disaster.[1] It begins by outlining the research objectives, background, and case studies' selection. Then, the research questions are presented, along with methods and data collection strategies. Finally, key limitations and ethical considerations are addressed, concluding with an overview of the thesis structure.

1.1. Background and Research Objectives

Mutual aid is a concept that has been around for centuries. Local communities have always been first responders when conflicts and disasters erupt. There are countless historical and contemporary examples showcasing how grassroots movements have stepped in during humanitarian crises, being far more coordinated and organised than what the formal humanitarian system is willing to acknowledge (Loy and Worley, 2025). However, it seems that the sector is experiencing a shift due to the increasing relevance of these neighbour-led initiatives, emphasized by international actors' and governments' lack of access, resources, and/or political will (Carstensen et al., 2021, p. 162).

In times of crisis, local communities often seek assistance from non-institutional actors due to proximity and familiarity motives. These actors include family members, neighbours, diaspora or religious groups, and in more recent times, digital communities (Hirono and Nurdin, 2024, p. 2). Conversely, the policy community and the humanitarian sector operate under the assumption that aid recipients primarily seek assistance from institutionalised actors, such as local governments or established NGOs (ibid).

Since the aid sector's localisation agenda has not materialised as quickly as expected, certain chronic illnesses of the humanitarian system remain untreated. Embracing emerging forms of mutual aid could help reshape the overburdened sector (Carstensen et al., 2021, p. 166). Thus, this paper intends to address a research gap found in existing mutual aid literature, drawing lessons from its contemporary applications in humanitarian crises. Few academic pieces have researched this phenomenon through an empirical lens.

[1] The term "natural disaster" is used to refer to the result of a combination of natural hazards and social/human vulnerability. Natural disasters do not solely happen because of natural factors; they are a consequence of socio-economic conditions – e.g. discrimination, inequitable access to resources, etc. (O'Keefe et al., 1976).

1.2. Case Study Selection: Sudan and Spain

Two primary case studies, Sudan and Spain, have been selected to carry out the empirical analysis of mutual aid responses. Sudan's locally led initiatives, known as Emergency Response Rooms (ERR), have emerged in the developing African country. On the other hand, Valencia's DANA floods in Spain exemplify volunteer responses taking place in a developed European country. The criterion behind these case study choices lies on their *relevance, context contrast, scalability,* and *subjective interest*.

In terms of relevance, Sudan's ERR have been a very popular phenomenon. Thriving in a conflict-affected environment characterized by limited resources and fractured governance, they have emerged as the most significant form of assistance, filling the vacuum left by formal humanitarian actors (Carstensen and Sebit, 2023). Conversely, Valencia's volunteer response after the DANA floods in October 2024 was very unexpected, resulting in community efforts almost replacing state mechanisms. This generated public discussions around the role of community-led responses in crisis (Martín, 2024).

When it comes to context contrast, the difference between the two cases is evident. While Spain is a stable country where disaster response mechanisms are institutionalized, Sudan suffers from a lack of democratic institutions and no social security net. Regardless, in both contexts there has been a need for mutual aid, challenging the systematic reliance on traditional humanitarian actors and government bodies in nowadays' crisis response strategies (Europa Press, 2024). Comparing these two cases allows us to acknowledge vastly different socio-political, economic, and infrastructural circumstances in diverse regions of the world.

Additionally, Spain and Sudan are cases that favour an exploration of mutual aid's scalability in different crises. Both examples have been depicted as successful and impactful mutual aid initiatives, providing strong criticisms regarding the role of external and formal actors concerning the overall humanitarian response (Olson et al., 2024; Monzón and Chaves, 2024).

Lastly, the proposal of Sudan and Spain is linked to a subjective interest in these cases motivated by the magnitude and gravity of Sudan's humanitarian crisis, as well as the fact that Spain is the author's country of origin. The choice was a difficult one, since the dynamics of mutual aid in different contexts remain under-researched.

For this reason, two secondary cases are explored to offer complementary perspectives, and avoid misconceptions when discussing local responses in developed vs. developing countries. Ukraine and Morocco's mutual aid initiatives are therefore selected following the same four criteria.[2] A response to armed conflict in a developed European country is compared to a disaster response in a developing African country. These cases are briefly discussed to increase the conclusions' validity and enrich the analysis.

1.3. Research Questions

To formulate the research question, the selected dependent variable is the impact of mutual aid initiatives, operationalised into three categories: (1) benefits, (2) limitations, and (3) long-term effects. The independent variable is the type of crisis setting, identifying armed conflict and natural disaster as the two classifications for the case studies. The research question is explanatory in nature, as it seeks to understand and explain how different crisis settings influence the impact of mutual aid initiatives, rather than evaluating their normative value.

The decision of focusing on the impact instead of on the emergence of mutual aid is based on the notion that locally led responses tend to be justified by a destabilization of formal structures in crises (Fechter and Schwittay, 2019, p. 1774). The type of setting where these unfold does not necessarily alter the emergence of mutual aid (Solnit, 2009, p. 2), but it might cause variation in its impact. To study the influence of mutual aid beyond external structures, a set of themes constituting a successful response are identified in the literature and explored in practical cases.

Research Question: How does the impact of mutual aid responses in contemporary humanitarian crises differ in cases of armed conflict (Sudan's Emergency Response Rooms, 2023-2025) vs. natural disaster (volunteer networks in the DANA floods in Spain 2024)?

Sub-question 1: Which are the benefits, limitations, and long-term effects determining the impact of mutual aid responses by grassroots organizations in the primary cases of Sudan's ERR and Valencia's volunteer networks after the floods?

[2] Relevance, context contrast, scalability, and subjective interest.

Sub-question 2: Which additional benefits, limitations, and long-term effects can be identified in the secondary cases of grassroots mutual aid in Ukraine's war (2022-2025) and Moroccan civil society's response after Al-Haouz's Earthquake (2023)?

These research questions focus on mutual aid's effects in different contexts. This is deemed to be the best approach considering the thesis' time constraints and depth elements. Impact is a more narrowly defined aspect of mutual aid, measurable through factors such as effectiveness, sustainability, and adaptability. Analyzing impact allows to offer recommendations, as well as assess whether mutual aid should be more integrated into formal humanitarian responses.

1.4. Methods and Data Collection

This research is a comparative study based on a qualitative analysis of primary and secondary data. Mutual aid's impact in Sudan and Spain is studied and contrasted, in search of common patterns and differences across contexts. The aim is to assess the benefits, limitations, and long-term effects of mutual aid using a Most Different Systems Design (MDSD) to isolate and control for other variables in markedly different case studies (Anckar, 2008).

A content analysis is conducted to identify core themes within the case studies. This research design allows to delve deeper into primary data through five semi-structured interviews, combined with a review of secondary sources tackling mutual aid's impact in Sudan and Spain. A coding framework displayed in manually drafted Excel thematic tables allows to organize findings into meaningful categories (Naeem et al., 2023), assigning code labels to statements/arguments, which facilitates the comparison of findings across cases.

1.5. Limitations and Ethical Considerations

The main limitation is the lack of literature on the topic of mutual aid's impact in armed conflict vs. natural disaster settings. There is a substantial volume of academic research in disaster sociology, but not in armed conflict sociology. Instances of comparative studies between both contexts have not been found. This is a challenge and, simultaneously, an opportunity to delve into an unexplored field of research, opening a door to what we could denominate the overlooked realm of 'Mutual Aid Studies'.

Finally, interviews have been conducted following the ethical principles of informed consent and confidentiality. Interviewees are fully informed about the purpose of the study and how their data will be used. Special care has been taken to anonymise their identities via the use of pseudonyms, ensuring that participation does not cause any additional burden. Semi-structured interviews have been selected as the chosen model to minimize risks of misinterpretation.

1.6. Thesis Outline

The thesis is structured into seven chapters. The first two introduce an explanatory view of the research, including the conceptual framework and methodology. Next, a literature review chapter delves into previous research and theoretical standards relevant for mutual aid's impact. The fourth and fifth chapters correspond to the case studies' display of Sudan and Spain's mutual aid data. The sixth chapter deals with the comparative discussion of findings, analyzed through thematic coding. Lastly, benefits, limitations, and long-term effects of community-led initiatives are critically assessed to produce consistent recommendations and conclusions.

2. Conceptual Framework and Methodology

This chapter presents the study's theoretical and methodological foundations. It delineates Mutual Aid Theory in crisis settings from a sociological perspective, combining it with insights from emergency management literature. This view places mutual aid within broader social processes, breaking down its impact into three constituent elements: benefits, limitations, and long-term effects. Finally, the methodological approach is detailed, including the comparative research design, data collection strategies, and the rationale behind the chosen analytical tools.

2.1. Conceptual Framework

The conceptual framework of this thesis adopts the combined perspective of Mutual Aid Theory by Kropotkin, with sociology studies on the emergence of collaborative adaptation in crisis situations (Spade, 2020a). In this section, theoretical perspectives on emergency management and grassroots interventions are linked to humanitarian action. By analyzing responses to crises from a community-based viewpoint, we understand the mutual aid institution as both a potential alternative, and a complementary humanitarian initiative for survival in modern-day emergencies.

2.1.1. Mutual Aid Theory

Pëtr Kropotkin's theory of Mutual Aid (1902) sustains that collective action and intra-group cooperation are natural factors of evolution that have historically determined the preservation of species. He defines the mutual aid institution as the origin of community's moral instincts, coming from a natural drive for survival. In his view, feelings of solidarity and mutual support go beyond love and personal sympathy, being rooted in an innate endurance impulse. Kropotkin (1902, p. 9-10) identifies the emergence of mutual aid as an instinct that has slowly developed among animals and humans alike during inherited evolutionary processes.

Mutual Aid Theory argues that nature's brutality does not result in Darwin's Theory of Natural Selection. Instead, Kropotkin (1902) presents arguments in favour of survival when communities follow a society-based lifestyle that enacts mutual support, claiming that "sociability is the greatest advantage in the struggle for life" (p. 57). While societies based on mutual aid are more likely to persist and progress, unsociable species are doomed to decline (ibid, p. 269).

In a contemporary context, Kropotkin's theory materializes into volunteering organizations that aim to preserve the life and well-being of communities, exemplifying the tendency of humans towards the creation of a global conscience of solidarity through unambitious and unpaid/underpaid work (Kropotkin, 1902, p. 259). Kropotkin's research also describes how mutual aid initiatives have been suppressed by centralized powers within prevailing systems of individualism for centuries. Still, he argues that these initiatives can be observed in everyday relations and, especially, within extraordinary circumstances (ibid, p. 214), guaranteeing survival for next generations.

Kropotkin claims that people generally feel compelled to help when witnessing appeals for aid, activating a psychological response. Once one person initiates the act of providing relief, a chain reaction occurs (Kropotkin, 1902, p. 254). He also identifies two different sets of intentions influencing mutual aid institutions from an ethical point of view. One is a self-interested motive, i.e. the desire of acquiring notoriety, political power, or social distinction. The other is the true feeling of human solidarity, deeply lodged in humans' understanding of societal norms. Regardless of which set of intentions is behind modern mutual aid institutions, they have reaffirmed themselves in scenarios of extraordinary need, defining civilizations' progress and survival (ibid, p. 275).

Kropotkin's Mutual Aid Theory (1902) allows us to identify four types of conditions defining the phenomenon's emergence: (1) social, (2) structural, (3) psychological, and (4) ethical. Kropotkin's view on the rise of mutual aid is interpreted from a sociological outlook by writer Rebecca Solnit (2009) in her work 'A Paradise Built in Hell', where she reviews Kropotkin's theory and contributes to the identification of these conditions.

- Social conditions should be characterized by pre-existing and strong community networks. People who are accustomed to inhabit close-knit societies believe that everyone is connected by a shared struggle (Solnit, 2009, p. 302). Therefore, crisis situations increase solidarity among populations sharing a common identity.
- Structural conditions must allow cooperation when triggered by an existential threat. Hierarchies and administrations tend to fall apart in crisis, delegating the disaster response to communities who have the chance to come together (ibid, p. 91). Hence, a lack or inadequacy of institutional or international responses gives way to dispersed and decentralized decision-making systems led by civil society (ibid, p. 305).

- Psychological conditions are required to showcase empathy, and predisposition for collective action. Crisis responses can bring moments of connection and moral clarity, creating a landscape of solidarity. A psychologically healthy society that follows intrinsic human tendencies and instincts described by Kropotkin's theory, will have individuals who help others in distress (ibid, p. 5).
- Ethical conditions might vary between altruistic intentions vs. strategic motivations, but they must be present to ignite spontaneous aid initiatives. Feelings of solidarity can become stronger during crisis, but opportunistic self-assertion behaviours can also push individuals to participate in community-led responses (ibid, p. 195).

2.1.2. Sociology of Survival in Crises

Glassman (2000, p. 392) summarizes Kropotkin's ideas from a sociological lens, arguing that the only way a species can thrive in a hostile environment is through extraordinary cooperation. Disasters and conflictual scenarios are factors that threaten the survival of populations and their human development, which triggers a reactive behaviour to natural and external checks. Resorting to forms of mutual aid and collective action to prevent the destruction of their communities is a common pattern for humans (ibid, p. 398).

From Glassman's (2000) and Kropotkin's (1902) perspectives, the lack of community represents the eradication of any joy of life, which bases the mutual aid instinct on the premise that the community's preservation overrides the preservation of the self (Glassman, 2000, p. 399). Mutual Aid Theory sees ethical behavior as a set of moral rules created by society to protect its integrity and long-lasting development (ibid, p. 409). When looking at contemporary understandings of mutual aid, Spade (2020a p. 131) argues that expanding the use of mutual aid in crises is the most effective way to support vulnerable populations so they can survive, mobilize, and overcome disasters

Spade's definition of mutual aid is the most adequate for this thesis's conceptual framework. He defines mutual aid as "a form of political participation in which people take responsibility for caring for one another, and change political conditions by building new social relations that are more survivable" (Spade, 2020a, p. 136). There is nothing new about mutual aid, but it has become more significant in crisis situations through the presence of collective care. Spade determines that effective social movements always include elements of mutual aid, regardless of the wider context (ibid).

Mutual aid often exposes failures in current systems and provides alternatives that counteract demobilization and individualism. It is a movement that goes further than volunteerism[3] but does not quite become formal humanitarian action.[4] By creating spaces where people come together based on shared needs/concerns, it promotes solidarity, participation, and unified decision-making (Spade, 2020a, p. 137). In these spaces, participants are not required to be 'professionals', which makes mutual aid an anti-authoritarian, non-hierarchical, decentralized, and democratic institution (ibid, p. 138). Spade also highlights how mutual aid should not be understood as charity[5], since it does not come with eligibility requirements related to moral frameworks of deservingness (ibid, p. 142).

2.1.3. Emergency Management in Disaster

When it comes to society's structure during emergency management, Stallings and Quarantelli (1985, p. 94) classify popular initiatives as 'emergent groups', highlighting their newness, absence of formalization, and lack of tradition. This implies that no public organization exists to respond to the situation, or that existing ones are incapable of doing so (ibid). These groups can be thought of as private citizens working together in pursuit of collective goals, without becoming institutionalized. The authors (ibid, p. 95) identify a horizontal hierarchy, a division of not highly specialized labour, and a significant percentage of volunteers as the main characteristics of these mutual aid groups. Many of them only exist for short periods of time, but recognition of their efforts seems to be an impetus for their continued existence.

Additionally, Stallings and Quarantelli (ibid, p. 98) argue that occurrences of mutual aid are inevitable, neither dysfunctional nor conflictive, and cannot be eliminated by previous planning. Emergence will happen either before, during, or after crises, stimulated by problems of interorganizational coordination, or because of demands excessing routine capabilities. The emergence of mutual aid groups shows that there are alternative possibilities to deal with crises beyond established systems (ibid).

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[3] Defined by Carstensen et al. (2021, p. 156) as "individuals or larger groups offering time, skills, and resources for the common good" resulting in cooperation among citizens, authorities, and NGOs in crisis.

[4] Institutionalized and professionalized system that offers humanitarian aid through established organizations, INGOs and local NGOs, following principles of neoliberal governance (Sandri, 2018, p. 66).

[5] A term denoting the provision of support for survival to poor and/or struggling people coming from richer sections of society or the government (Spade, 2020a, p. 140)

2.1.4. Grassroots Interventions in Humanitarian Action

Fechter and Schwittay (2019, p. 1769) utilize the term ‘citizen aid’ to refer to mutual aid initiatives instigated by individuals/communities to support those in need. Other authors like Taithe (2019), and Sandri (2018) emphasize the agency of local, non-professional individuals providing relief in crisis situations, labeling it as ‘demotic’ or ‘volunteer’ humanitarianism. These forms of intervention are small, temporarily limited, and usually remain under the radar (Fechter and Schwittay, 2019, p. 1770). With grassroots humanitarianism, conflicts and natural disasters become sites reflecting shared humanity, social justice, and horizontality. Sadly, these movements are often dismissed due to their ‘amateurish’ nature (ibid, p. 1773), not falling under the category of ‘local NGO movements’ involving new volunteers, but resulting in privately funded initiatives sustained by personal transnational networks.

2.1.5. Operationalization of Mutual Aid’s Impact

To assess the impact of mutual aid in the selected case studies, it is fundamental to conceptualize the three previously mentioned categories: benefits, limitations, and long-term effects.

When analyzing at the benefits of mutual aid in crises, the author defines them as ‘positive and constructive key factors that maximize the impact of community-led initiatives.’ These factors are differentiators from formal humanitarian responses. The main emergent themes in this category are community involvement, speedy and flexible responses, localized knowledge, and development of solidarity networks (Hirono and Nurdin, 2024; Rose and Adler, 2024; Spade, 2020b; Knearem et al., 2023; Kotani and Yokomatsu, 2016; Solnit, 2009).

Moving to mutual aid’s limitations in crises, they are framed as ‘negative and adverse key factors that reduce the impact of community-led initiatives.’ These factors are differentiators from formal humanitarian responses. The main identified themes in this category are lack of logistical coordination, legitimacy challenges, resource scarcity, and safety threats (Carstensen et al., 2021; Soden and Owen, 2021; Spade, 2020b; Schenkenberg, 2016; Solnit, 2009).

Lastly, long-term effects of mutual aid in crisis imply 'lasting consequences of positive or negative nature, generated by the impact of locally led responses that produce durable changes in the structures and behaviours of affected communities.' The typical legacy of mutual aid efforts includes trust and capacity building, civic engagement, higher accountability and transparency, and governance shifts (Kynoch, 2025; L2GP, 2024; Spade, 2020b; Knearem et al., 2023; Wang, 2013).

The combination of these three operationalized categories allows a detailed analysis of mutual aid's impact in the empirical cases of Sudan's ERR and Spain's DANA volunteer groups, as well as those of Ukraine and Morocco's mutual aid initiatives.

2.2. Methodology

To answer the main research question and two sub-questions, a threefold methodological approach is pursued to conduct a comprehensive analysis of mutual aid's impact in armed conflict vs. natural disaster settings. Firstly, a literature review establishes the theoretical standards of mutual aid in the field of humanitarian action. Secondly, qualitative research is carried out by using primary and secondary data to analyze mutual aid's impact in Sudan and Spain, including insights from interviews with key stakeholders. Thirdly, the data is examined using thematic analysis to identify common patterns and differences.

2.2.1. Analytical Framework

The research is based on a mixed methods approach, adopting a comparative analysis that reflects on qualitative data collected from two main case studies. Empirical findings are employed to explore Sudan's ERR and Valencia's volunteer groups in depth. Information is collected through both primary (interviews and first-hand records) and secondary sources (NGO reports, academic articles, news publications, etc.). This data is complemented with the brief analysis of secondary sources from Ukraine and Morocco's additional case studies.

Mutual Aid Theory by Kropotkin is therefore used to conceptualize the impact dynamics and differences/similarities between the two contexts, while observing practical consequences of community-led initiatives for humanitarian action. This thesis also

draws from the perspective of 'Critical Theory' popularized by the *Frankfurt School*[6] (Celikates and Flynn, 2023) to critically analyze the implications of mutual aid for humanitarian action.

The present analytical framework entertains multiple hypotheses, acknowledging that there could be competing findings depending on the case selection and theoretical approaches chosen (Robson and McCartan, 2016, p. 67). The research adopts a primarily inductive logic, seeking to construct general insights from empirical observations, as opposed to testing pre-established theoretical hypotheses (Anckar, 2008).

However, it is important to note that the assessment of the dependent variable, the impact of mutual aid in different contexts, is mostly speculative and follows an interpretative methodology of the data available so far. The character of this research does not allow to present definitive arguments about long-term effects of mutual aid in still ongoing crises or early recovery stages.

2.2.2. Methodological Framework

Following the logic of a MDSD, the comparative analysis is based on the analogy of case studies with contrary independent variables, applicable to both instances of Sudan vs. Spain, and Ukraine vs. Morocco. This approach intends to highlight how mutual aid functions in vastly different contexts, acknowledging the variety of structural and contextual factors while simultaneously trying to identify recurring patterns across the case studies' mutual aid initiatives (Anckar, 2008).

The data is analyzing through Thematic Analysis (TA), a qualitative method that allows a structured comparison of the benefits, limitations, and long-term effects of mutual aid in natural disaster vs. armed conflict settings. The principles of this technique include the coding of data and searching for themes to identify patterns in a dataset, which are then interpreted for their inherent meaning. These can be detected based on the extraction of key concepts from selected sources (Naeem et al., 2023, p. 2).

[6] Framework that aims at critiquing and transforming society by integrating normative perspectives with empirically informed analysis of society's conflicts, contradictions, and tendencies (Celikates and Flynn, 2023).

Each section is supported by coded excerpts from sources presented in chapters four and five and then, gathered in manually elaborated Excel tables summarizing key trends.[7] Tables are used as a way of displaying textual data to make systematic comparisons between Sudan and Spain in a standardized manner (Robson and McCartan, 2016, p. 477).

2.2.3. Data Collection and Sources

It is important to acknowledge a slight unbalance regarding data quality, collection methods, and sources' substance between the two main case studies. Given that the author has a closer personal and professional link to Spain, the collection of data for this case has been smoother. For Sudan's case, language and access barriers have been considerable setbacks. However, these aspects have not undermined the validity of the research's outcomes.

The primary data of the case study chapters consists of five semi-structured interviews. Two of them are subjects of Sudan's case study, and the other three correspond to Spain's case study. A guiding structure for the questions[8] was used, but the answers' format was substantially modified and adapted based on the flow of the interview (Robson and McCartan, 2016, p. 285). Interviewees, who have been anonymized, are volunteers, humanitarian workers, community members, students, journalists… All of them relevant stakeholders with first-person insights on the impact of mutual aid initiatives in Sudan and Spain.

Secondary data is extracted from a content analysis of published reports, media articles, and academic literature covering mutual aid efforts in the four chosen case studies. A re-analysis of case study reports provides complementary and contrasted views to primary data (Robson and McCartan, 2016, p. 359) particularly relevant in the cases of Sudan and Spain. This allows a critical assessment of the impact of informal response mechanisms in humanitarian crisis scenarios.

[7] Appendices 1 & 2 show the tables coded according to the TA framework, displaying key statements and arguments, useful concepts, codes, and helpful themes for the conceptualization of mutual aid's impact in the two primary case studies.

[8] See Appendix 3 for the questions included in the standard interview structure used with the five relevant stakeholders.

3. Literature Review of Mutual Aid in Humanitarian Action

This chapter provides an overview of academic literature concerning mutual aid, analyzing the elements of its impact. The goal is to shed light on the role of grassroots interventions in conflict and disaster settings, showcasing the theoretical standards later identified and applied in the case studies' analysis.

3.1. Previous Research

Mutual aid is inherent to a lot of survival stories, and it showcases the reliance in some form of community networks when crisis happen (Steinberg, 2003, p. 31). The notion that individuals can assist one another when navigating adversities is recurring in the discipline of sociology, which makes mutual aid both a utopia and a reality (Dynes, 2006, p. 10). In the long run, mutual aid encompasses beneficial emotional, intellectual, and material dimensions, informing societies' risk preparedness and resilience (Steinberg, 2003, p. 38). Such conceptualization aligns with views in critical humanitarian discourse that emphasize the agency of affected communities, often undermined by top-down humanitarian practices (Brun and Horst, 2023, p. 63).

However, Solnit (2009, p. 8) cautions about the assumption that spontaneous volunteer groups emerge exclusively because of inadequacy or absence of official responses. Although we could assume that mutual aid efforts just exist to 'fill the gap' until they are eclipsed by official relief agencies, they should be perceived as common localized survival strategies and valid humanitarian responses (ibid, p. 294).

An interesting phenomenon characterizing mutual aid in crisis is what Charles Fritz labels as *convergence,* a massive movement of people and volunteers toward a disaster area, usually more significant than evacuations away from impacted regions (Solnit, 2009, p. 195). The contrary happens when war erupts, with many more people fleeing than staying behind to participate in grassroots efforts (Schenkenberg, 2016, p. 19). Knowing this, little attention has been given to the essential distinction between humanitarian conditions in natural disasters compared to armed conflicts, especially in terms of the localization agenda. Hence, the role of local actors in natural disasters is much less controversial and challenging than in armed conflicts (ibid, p. 23).

3.1.1. Contemporary Mutual Aid

Kropotkin's theory applied by scholars such as Solnit (2009) reveals that mutual aid is a default operating principle of civil society when disaster and conflict strike. Thus, crisis contexts do not result in a Hobbesian 'law of the jungle' scenario, but in a kind of anarchy where citizens organize effectively, building capacities to create more sustainable post-crisis societies (Spade, 2020b, p. 73). Solnit (2009, p. 226) recognizes the value in local responses, identifying them as a form of self-determination pointing to tensions between institutional humanitarianism and bottom-up aid (Brun and Horst, 2023, p. 64).

Lastly, it is relevant to mention the role of mass media as a new factor in mutual aid responses. Digitalization's impact has shaped grassroots coordination in both Sudan and Spain, and in previous instances like the COVID-19 pandemic (Soden and Owen, 2021, p. 3). A never-seen-before involvement in mutual aid digital networks has sparked a wide range of new elements for effective self-organisation, which are being studied to assess the impact of new technologies in crisis management (ibid). As a result, online platforms have been increasingly used as channels for humanitarian responses to mobilize, spread the word about needs, and connect faster across geographic boundaries (Kynoch, 2025).

3.2. Theoretical Standards of Mutual Aid's Impact

To operationalize the three categories of mutual aid's impact, a selection of academic articles, books, and NGO reports are used to craft this literature section. Common themes are identified, which will be implemented afterwards as part of the discussion chapter.

3.2.1. Mutual Aid's Benefits in Literature

According to experts, the benefits of mutual aid's impact include four main themes, usually not present in formal humanitarian responses. These are common in all types of crisis settings, both conflict and disaster.

- Political mobilization and collective action. Mutual aid is key to social movement formation in humanitarian crises, but also within daily-life conditions. Individuals join groups seeking specific support, which usually implies a political demand/revindication. The ability to access assistance in times of crisis is a prerequisite for political activity, given the challenges of mobilization when facing existential threats (Spade, 2020b, p. 9).

- Building solidarity. The society that Kropotkin dreamed of attempts to render altruism and care as permanent and strengthened community pillars, going beyond emergencies (Solnit, 2009, p. 180). Since crises induce temporary widespread humanitarianism, human networks become more integrated. Consequently, society increases its network density, resulting in a more active social capital (Kotani and Yokomatsu, 2016, p. 330).
- Local participation and knowledge. Although the humanitarian sector's localization agenda tries to ensure cultural understanding of communities involved in their initiatives, local participation remains limited (Rose and Adler, 2024, p. 5). Mutual aid demonstrates that local knowledge has great potential to save lives, especially regarding expertise about the field in which first responders operate. Community members also relate better to each other and possess contextual appropriateness, catering to the community's unique needs more effectively and sustainably (Hirono and Nurdin, 2024, p. 6).
- Rapid responses. Since affected communities do not have to relocate to reach the victims, mutual aid responses are presumably always faster (Knearem et al., 2023). Timeliness of the response can often save more lives than specialized skills (Solnit, 2009, p. 310), so community-led initiatives are more favorable to guarantee rapidness. Furthermore, crises tend to overload political systems by multiplying societal demands, which reveals administrative deficiencies ingrained in bureaucracies. These render systems unsuccessful at addressing urgent needs due to lack of improvisation capacities and sluggishness (ibid, p. 152).

3.2.2. Mutual Aid's Limitations in Literature

According to specialists, the limitations of mutual aid's impact include four main themes, sometimes also present in formal humanitarian responses. These seem to be shared elements in all crisis settings, with varying degrees of incidence.[9]

[9] Literature identifies more limitations to mutual aid in armed conflicts due to higher uncertainty and insecurity. Community-led initiatives appear to struggle to bring humanitarian assistance to scale and deliver enough aid to affected communities within war settings (Schenkenberg, 2016, p. 5).

- Lack of logistical coordination. Clarity and task distribution can sometimes become a problem if activities of different local groups overlap (Schenkenberg, 2016, p. 8). In crisis situations, it is difficult for grassroots associations to establish organized structures because they struggle to create definite decision-making methods, sometimes resulting in participants not knowing exactly what is ensuing (Spade, 2020b, p. 34).
- Legitimacy challenges. There is a great difference between the perception of formal vs. informal humanitarian responses. When dealing non-professional initiatives, questions about expertise, validity, and adequacy are common (Soden and Owen, 2021, p. 14). Hence, mutual aid projects can be framed as threatening or inefficient efforts from the perspective of existing institutions, lacking recognition from authorities and established humanitarian actors (Solnit, 2009, p. 199).
- Scarcity and competition. Insufficient financial resources, time, and material capacity characterize mutual aid's outcomes. This scarcity leads to volunteer groups becoming competitive between and amongst themselves, which can create conflicts and miscalculations (Spade, 2020b, p. 34). Moreover, the scale of devastation can sometimes imply that mutual aid is not enough, exposing the need of professionalized assistance channelled through formal actors (Carstensen et al., 2021, p. 166).
- Safety threats. Groups carrying out frontline mutual aid in emergencies face particularly risky conditions. They must navigate threats due to their exposure to natural hazards and open fire without any professional protection (Spade, 2020b, p. 18). Therefore, volunteers' lives are constantly in danger, which ultimately impacts the scale and quality of the assistance received by the people in need (Schenkenberg, 2016, p. 3).

3.2.3. Mutual Aid's Long-term Effects in Literature

Despite mutual aid tending to have mostly short-term effects, there are factors that enable its long-lasting impact. As power structures regain control post-crisis, mutual aid's impact can be co-opted or even completely erased (Spade, 2020b, p. 28). Four main themes are identified:

- Trust-building. Community responses generate a sense of belonging and reliance within affected societies. This can bring elevated trust and cohesion, since mutual aid is characterized by reciprocity and absence of indebtedness forces, balancing interpersonal ties (Knearem et al., 2023, p. 590). Effective intergroup coordination can also raise questions about the role of governments, possibly creating distrust towards established structures proven to be inadequate (Spade, 2020b, p. 20).
- Accountability and transparency. Locally led efforts are closer to beneficiaries, allowing for greater feedback intake. Direct interaction and localized operations also facilitate better monitoring/evaluation processes (Wang, 2013, p. 427), enabling self-organising groups to quickly formulate and implement their own responses. This results in retained accountability, compliance, and cost-effectiveness (L2GP, 2024).
- Enhanced resilience and empowerment. Mutual aid can reduce dependency on external actors and hierarchical structures between recipients/providers, enhancing affected communities' agency and creating sustainable support systems (Knearem et al., 2023, p. 566). In the long-term, mutual aid can create a sense of counterrevolution, showcasing enhanced resilience and civil society empowerment (Solnit, 2009, p. 153).
- Greater autonomy. Lesser reliance on governmental institutions when crisis arise can be a long-term product of mutual aid. Although a strong and united civil society cannot prevent disasters, it can plan and prepare accordingly for them in future occasions (Solnit, 2009, p. 266). Additionally, beyond immediate relief, mutual aid networks can serve as a foundation for lasting self-sufficiency and collective problem-solving expertise (Kynoch, 2025).

All twelve themes extracted from mutual aid's benefits, limitations, and long-term effects will be identified and analyzed through the case studies' data in the following three chapters.

4. Sudan Case Study: ERR and Mutual Aid Coalitions

This first case study tackles contemporary mutual aid initiatives in armed conflict settings, describing their emergence and impact. Data is retrieved from both primary and secondary sources, including interviews to relevant stakeholders involved in Sudan's ERR and mutual aid efforts. Two people were interviewed in online meetings: an OCHA worker, and a Sudanese journalist and ERR volunteer. Additionally, NGO and CSO reports, academic publications, and news articles covering Sudanese mutual aid responses are used to find consistencies and discrepancies relating to the standards presented in the literature.

4.1. Context and Emergence

When Sudan's civil war began in April 2023, the confrontation between the Rapid Support Forces (RSF) and the Sudanese Armed Forces (SAF) brought destruction, loss, and mass displacement to the country (Kuka, 2024). Given the difficulties to access conflict-affected areas, the collapse of the state, and a slow-moving international relief effort, youth-driven volunteer networks set up ERR in response to the humanitarian crisis (Nasir et al., 2023). Despite the magnitude of threats and challenges for Sudanese civilians, the number of acts of solidarity surpassed institutional humanitarian efforts (Carstensen and Sebit, 2023).

Community-based mutual aid groups continue to play a leading role in the humanitarian response to the conflict in Sudan, as security issues prevent international relief agencies from accessing affected areas (Nasir et al., 2023). ERR have been built with the same spirit that toppled al-Bashir's regime, following values based on four pillars of good governance: accountability, transparency, participation, and equality (Kuka, 2024).

4.2. Mutual Aid's Impact within Armed Conflicts

To observe the impact of mutual aid efforts in Sudan's armed conflict context, excerpts of primary data (interviews) and secondary data (reports, publications, news pieces, etc.) have been coded in an excel table[10] following the methodology of TA. The main findings are presented in a summarized table below, showing insights regarding the benefits, limitations, and long-term effects of mutual aid's impact.

[10] See Appendix 1.

Mutual Aid Impact Categories	Themes	Case Study Insights
Benefits	1. Political mobilization and collective action 2. Building solidarity 3. Local participation and knowledge 4. Rapid responses	ERR have presence in remote areas, allowing rapid mobilization and equal distribution of services to communities (OCHA worker, 2025). They avoid bureaucratic barriers and foster a participative model of aid (Sudanese journalist and volunteer, 2025). Volunteers understand local dynamics, favouring a humanitarian response that adopts a bottom-up approach (OCHA worker, 2025). Psychosocial support and collective care have also been crucial elements to their initiatives (Sudanese journalist and volunteer, 2025).
Limitations	5. Lack of logistical coordination 6. Legitimacy challenges 7. Scarcity and competition 8. Safety threats	Volunteers have faced threats and attacks from armed groups (OCHA worker, 2025; Kleinfeld, 2025). Security risks in warzone areas are extremely high, and the lack of resources and funding constraints limit the impact of mutual aid initiatives (Olson et al., 2024). Some formal actors, as well as the parties involved in the conflict, have refused to recognize ERR and allow activities in their controlled areas due to fear of corruption and mistrust (Sudanese journalist and volunteer, 2025).
Long-term effects	9. Trust-building 10. Accountability and transparency 11. Enhanced resilience and empowerment 12. Greater autonomy	Communities are demonstrating they can be accountable and entitled to build their own systems, protecting and taking care of each other (Sudanese journalist and volunteer, 2025). Volunteers and activists are building capacities, knowledge, and becoming more empowered and resilient to respond effectively in future crises without relying on external actors (Carstensen and Sebit, 2023). ERR have introduced a new form of decentralised service-delivery that provides greater autonomy (Nasir et al., 2023).

Table 1. *Mutual Aid's Impact in Sudan's Armed Conflict (Source: Author's own, 2025).*

4.3. Parallels with Ukraine's Mutual Aid Response

To support the findings of Sudan's case study, similarities concerning the impact of mutual aid have been researched in another armed conflict setting: Ukraine's war prompted by Russia's invasion (2022–). According to relevant pieces of literature, the themes of mutual aid's impact categories described in the previous section are mostly analogous.

The benefits of mutual aid initiatives found in the Ukrainian case show strengthened social cohesion, political mobilization, and participation of civilians in humanitarian efforts (Madryha et al., 2024, p. 189). New feelings of solidarity and connectedness among Ukrainians from different regions emerged, with informal networks filling aid gaps (Deineko, 2023, p. 171-2).

War-related limitations for mutual aid initiatives in Ukraine include security threats, access barriers to conflict zones, lack of coordination, and resource constraints. The provision of humanitarian assistance under life-threatening conditions, as it happens in the case of Sudan, affects the nature and specifics of mutual aid, making it a very dangerous and unpredictable task (Madryha et al., 2024, p. 192).

Lastly, long-term effects of mutual aid in the case of Ukraine have contributed to a significant strengthening of the societal cohesion during the armed conflict. Deineko (2023, p. 173) argues that evidence from the war shows a development of mechanisms to obtain greater autonomy and implement bottom-up initiatives. This generates an 'inclusive identity' that has the potential to become a core element for empowerment and nation-building in post-war phases.

5. Spain Case Study: Valencia's Floods and DANA Volunteers

This second case study tackles contemporary mutual aid initiatives in natural disaster settings, describing their emergence and impact. Data is retrieved from both primary and secondary sources, including interviews to relevant stakeholders involved in Valencia's volunteer groups and flood-affected communities. Three people were interviewed in online and in-person meetings: a mutual aid organizer, a Spanish community member, and a student volunteer. Additionally, NGO and CSO reports, academic publications, and news articles covering Spanish mutual aid responses are used to find consistencies and discrepancies relating to the standards presented in the literature.

5.1. Context and Emergence

After the DANA floods struck Valencia on October 29, 2024, vulnerabilities in local infrastructures and emergency preparedness protocols were exposed (Martín, 2024). In response, Spanish civil society rapidly mobilized tens of thousands of volunteers to coordinate aid distribution on the ground, relying on traditional solidarity networks and neighbourhood associations, who took the lead in organizing relief efforts (Monzón and Chaves, 2024).

During the first days after the disaster, grassroots organizations were the main actors providing aid, carrying out rescue missions, and sending cleaning brigades to the most affected areas (Europa Press, 2024). Online platforms and digital mutual aid hubs such as 'Ayuda Terreta' and 'Xarxa Suport Mutu DANA Valencia' were created to collect demands, diagnose needs, and provide essential services (Saiz Saéz, 2024). Institutional responses were activated shortly after, but seven months later, Valencia continues to recover. Collaboration between civil society and governmental institutions remains crucial in addressing the disaster's long-term effects (Collado and Benlloch, 2025).

5.2. Mutual Aid's Impact within Natural Disasters

To observe the impact of mutual aid efforts in Spain's natural disaster context, excerpts of primary data (interviews) and secondary data (reports, publications, news pieces, etc.) have been coded in an excel table[11] following the methodology of TA. The main findings are presented in a summarized table below, showing insights regarding the benefits, limitations, and long-term effects of mutual aid's impact.

[11] See Appendix 2.

Mutual Aid Impact Categories	Themes	Case Study Insights
Benefits	1. Political mobilization and collective action 2. Building solidarity 3. Local participation and knowledge 4. Rapid responses	Neighbours helping each other was the easiest way to work fast and in proximity with affected communities (Mutual aid organizer, 2025). Solidarity and mutual support among people were key to address the situation and quickly adapt to post-disaster challenges (Europa Press, 2024). Locals were the first responders to the crisis, coordinating communities, fostering solidarity, and creating temporary aid points (Spanish community member, 2025; Student volunteer, 2025).
Limitations	5. Lack of logistical coordination 6. Legitimacy challenges 7. Scarcity and competition 8. Safety threats	The Valencian government limited the movement and access of volunteers to the most affected municipalities the week after the disaster, creating bureaucratic restrictions and framing mutual aid efforts as competing services for state-led relief (Aliño, 2024). There was a clear lack of resources, capacity, and mobility in volunteer groups, as well as an absence of preparation or knowledge on how to act in emergency situations (Mutual aid organizer, 2025).
Long-term effects	9. Trust-building 10. Accountability and transparency 11. Enhanced resilience and empowerment 12. Greater autonomy	The main lesson learned from the floods is that neighbours should be politically organized and active in their communities, enhancing empowerment and creating mechanisms to facilitate coordination and resilience (Mutual aid organizer, 2025). Many local organizations are supporting reconstruction strategies in the affected areas, implementing bottom-up approaches and improving collective consciousness and participation (Collado and Benlloch, 2025).

Table 2. *Mutual Aid's Impact in Spain's DANA Floods (Source: Author's own, 2025).*

5.3. Parallels with Al-Haouz's Earthquake Local Response

To support the findings of Spain's case study, similarities concerning the impact of mutual aid have been researched in another natural disaster setting: Al-Haouz (Morocco) 7.2 magnitude earthquake on September 8, 2023. According to relevant pieces of literature, the themes of mutual aid's impact categories described in the previous section are mostly akin.

The benefits of mutual aid initiatives found in the Moroccan case show how civil society responded immediately after the earthquake, before public authorities could plan a relief strategy (Boukhris, 2023). Diaspora organizations mobilized through digital platforms, together with CSOs with local knowledge and presence, which facilitated coordination and provided a balanced, speedy, and effective distribution of support (ibid). Ordinary Moroccans showed impressive solidarity, mostly out of necessity due to the overwhelming absence of institutional leadership (Errazzouki, 2024, p. 201).

Disaster-related limitations for mutual aid initiatives in Morocco include the constraints put in place by the government, blocking volunteers from distributing mutually collected aid because they lacked proper authorisation (ibid, p. 202). Furthermore, the affected region already suffered from a lack of resources and infrastructure, challenging the abilities of grassroots groups to provide sufficient aid (ibid, p. 197).

Finally, the long-term effects of mutual aid in the case of Al-Haouz earthquake contribute to the collective call for more accountability and transparency, demanding the end of neglect and marginalization in the forgotten region of Al-Haouz (Boukhris, 2023). In the aftermath, local communities led rebuilding efforts with a sense of renewed resilience and empowerment to reimagine improved neighbourhoods (Errazzouki, 2024, p. 203).

6. Comparative Analysis and Discussion

This chapter compares the impact of mutual aid initiatives in Sudan (conflict) and Spain (disaster). The benefits, limitations, and long-term effects determining impact in the cases of Sudanese ERR and volunteer groups in Valencia are revisited to answer the main research question and first sub-question. Additional perspectives extracted from the secondary cases of Ukraine and Morocco are included to answer the second sub-question and validate the main case studies' findings.

Kropotkin's Mutual Aid Theory, combined with the sociological outlook presented by Glassman (2000), Spade (2020a), and Solnit (2009), is helpful in this comparative analysis as a framework that structures and provides greater nuance to the assessment of mutual aid in practice. The qualitative analysis of the data follows a threefold methodology, operationalizing the dependent variable through standards identified in the literature, and posteriorly codifying primary and secondary material following TA. This allows to identify patterns and inform recommendations presented in the conclusion.

6.1. Benefits of Mutual Aid in Conflict vs. Disaster

Looking into the benefits of mutual aid, political mobilization and collective action seem to be easier to materialize in natural disaster settings, as there are lesser dangers and numbers of people fleeing affected zones. This aspect does not vary when looking at states with more developed governance structures vs. developing ones. In post-disaster situations, a higher percentage of people choose to stay behind to provide humanitarian assistance and participate in reconstruction efforts (Spanish community member, 2025), and even outsiders mobilize to help – as in Valencia's case. In Morocco, citizens also had to take matters into their own hands due to authorities' inaction and a lack of international relief reaching affected communities (Errazzouki, 2024), common factors in both natural disaster contexts. In conflict settings, the incidence of civil society mobilization can be reduced due to exacerbated socio-political cleavages. The dilemma of mutual aid platforms remaining neutral – ERR, for instance – vs. adopting a political stance siding with one of the conflict's parties – Ukraine's case – can sometimes undermine the positive aspects of collective action (OCHA worker, 2025).

The benefit of building solidarity in crisis-affected communities through mutual aid efforts is a common factor in both armed conflict and natural disaster settings. Data from Spain and Sudan shows that increased social cohesion, collective care, and generosity are intrinsic characteristics of mutual aid efforts (Olson et al., 2024) regardless of the type of crisis they emerge in. Insights obtained from the two main cases, and supported by the two additional ones, indicate that people tend to come together when crisis happen. Furthermore, they show even more extraordinary signs of altruism the graver the needs are, and when there is a lack of action/support from institutions or external humanitarian organizations (Carstensen and Sebit, 2023).

Local participation and knowledge are present in similar ways in both conflict and disaster settings, but for different reasons. In Sudan's war-torn areas, it is very difficult to get external assistance due to access barriers and insecurity conditions. Therefore, local participation is key (Sudanese journalist and volunteer, 2025). Conversely, in post-disaster situations, there is increased legitimacy, familiarity, and certainty linked to locally led responses. This can be attributed to the fact that aid providers are, at the same time, members of the affected communities who fully understand the needs and struggles of beneficiaries (Martín, 2023). This applies to both natural disaster and armed conflict settings, where mutual aid shows that communities are always first responders and often the most effective coordinators of humanitarian assistance in all types of crises.

The last benefit, common and very remarkable in Sudan and Spain's mutual aid cases, is the rapid character of the responses. Grassroots agility in Sudan is attributed to the ERR's informal structure and lack of bureaucracy constraints, added to the ability of CSOs to have presence in remote areas where humanitarian organizations and governmental actors cannot reach (OCHA worker, 2025). In the case of Spain, neighbours helping each other was the easiest way to work in proximity with affected communities. Localized responses were much faster than the government's action, and volunteers delivered key assistance in the immediate aftermath of the DANA floods (Mutual aid organizer, 2025).

Overall, there appears to be a higher prevalence of mutual aid benefits in natural disaster settings. The findings of Spain's floods identify more beneficial aspects to mutual aid than those of Sudan's civil war. The same result is obtained when analyzing Ukraine and Morocco's community-led initiatives. Natural disaster settings prove to escalate advantages of mutual aid's impact to a greater degree than armed conflict contexts do.

6.2. Limitations of Mutual Aid in Conflict vs. Disaster

When analyzing the limitations of mutual aid's impact within the main case studies, the lack of logistical coordination is identified as one of the main challenges of grassroots initiatives. Since most organizations do not have previous experience, improvisation is one of the main strategies used, especially at the beginning of crises (Mutual aid organizer, 2025). Overlaps and communication difficulties constrain the coordination between community-based organisations, ERR, and mutual aid volunteers in Sudan (Carstensen and Sebit, 2023), and in Spain, volunteer networks had to eventually cooperate with formal actors because they did not have enough capacity to deliver the aid needed (Spanish community member, 2025).

Another barrier undermining mutual aid's impact is a set of legitimacy challenges, mainly coming from governments. If there is no recognition of grassroots efforts, their impact is reduced by bureaucratic/legal barriers. Spanish institutions deliberately ignored many of the appeals by mutual aid networks, avoiding communication with local groups and delegitimizing their actions (Mutual aid organizer, 2025), which also happened in the aftermath of Al-Haouz earthquake (Errazzouki, 2024). In Sudan, INGOs such as the UN quickly recognized the impact of ERR and mutual aid platforms (OCHA worker, 2025). However, confronted parties and designated authorities still have not granted them permission to be active in certain areas (Sudanese journalist and volunteer, 2025).

Scarcity and competition are also major problems for mutual aid initiatives, which deal with a permanent lack of resources. In Sudan, there is an evident funding shortage, which leads to inevitable competition among mutual aid groups over who can get more support and the means to continue with their mission (OCHA worker, 2025). Conversely, in Valencia the resources were in the government's hands, but since civilians could not access them, their actions were limited to what they had available (Martín, 2023). The secondary cases of Ukraine and Morocco mirror the same dynamics, with major shortages due to government's and third parties' lack of capacity or willingness to provide resources (Madryha et al., 2024; Errazzouki, 2024).

Additionally, safety threats are significantly more prevalent in armed conflict than in natural disaster settings. Security and access issues in Sudan have determined the impact of mutual aid efforts since their establishment and during the war's evolution (Nasir et

al., 2023). In Ukraine, insecurity deters individuals and groups from remaining in war-torn areas (Madryha et al., 2024, p. 192). This, together with access issues, is a common trend in all conflicts regardless of the geographical region where they take place. The cases of the DANA floods and Al-Haouz earthquake vary significantly in this aspect, since the access problem is a commonality, but the safety element is not. Due to the blockage of paths and roads by mud right after the storm, access to towns surrounding Valencia was very problematic, but once the rain alert was over, there were no life-threatening factors involved in locally led responses (Monzón and Chaves, 2024).

Overall, there appears to be a higher prevalence of mutual aid limitations in armed conflict settings. The findings of Sudan's civil war identify more detrimental aspects to mutual aid than those of Spain's floods. The same result is obtained when analyzing Ukraine and Morocco's community-led initiatives. Armed conflict settings prove to weaken mutual aid's impact to a greater degree than natural disaster contexts do.

6.3. Long-term Effects of Mutual Aid in Conflict vs. Disaster

To pinpoint the long-term effects of mutual aid, trust-building is one of the key themes in both contexts. In the case of Sudan, awareness-raising is one of the main goals of ERR. They offer workshops and mentorship programs aimed at providing volunteers with tools that embody trust-based relationships with colleagues and beneficiaries (ERR Khartoum State, 2025). Similarly, the Mutual aid organizer of 'Xarxa Suport Mutu Dana Valencia' (2025) highlighted the need to remain organized and active in our communities, building trust and creating mechanisms that can be quickly activated in case of emergency.

Moreover, accountability and transparency become essential elements of the mutual aid phenomenon in the long run. When the Valencian government did not provide the required response, volunteers were the ones who jumped in despite the extremely challenging circumstances. Now, in the reconstruction phase, mutual aid groups have created reconstruction committees to ensure accountability from authorities and make sure prevention mechanisms are put in place (Spanish community member, 2025). Alternatively, in the case of Sudan's ERR, transparency towards beneficiaries is achieved by involving them in all sorts of mutual aid efforts without relying on the state, ensuring the desired quality of services, and hopefully leading to a participative reconstruction process (Sudanese journalist and volunteer, 2025).

Enhanced resilience and empowerment are also some of the effects associated with mutual aid movements. Sudanese volunteers and activists are currently building their capacities and knowledge to respond more effectively to the current crisis, but they are also becoming more resilient in case of facing similar crises in the future (Carstensen and Sebit, 2023). The same thing applies to Spanish volunteers, specially those supporting reconstruction and bottom-up strategies in affected areas. Since the emergency is over, locally led platforms are empowered in the reconstruction process, rebuilding community and economic networks themselves (Collado and Benlloch, 2025).

To conclude, the theme of civil society's greater autonomy is also discussed in both primary case studies and seen in the findings of Ukraine and Morocco as well. Mutual aid is mostly understood as a complementary response in the case of the DANA floods; but it has become a pseudo-alternative to governance in Sudan's civil war. This seems to be attributable to the different dynamics between civil society and formal authorities in each country. While Spanish institutions are very stable, they created a sense of distrust and abandonment due to their unpreparedness to provide an emergency response (Spanish community member, 2025), leading to the belief that people could only rely on each other.[12] Conversely, the crushed Sudanese institutional structure forced communities to build their own systems from the start, relying on their networks without expecting anything from the warring parties (Carstensen and Sebit, 2023).

Overall, long-term effects of mutual aid appear to be the most uncertain and context-dependent of all categories. The findings of Sudan's civil war identify a greater potential for long-term effects of mutual aid because of the initiatives' prolonged existence and activity. The political and democratizing implications linked to ERR's project, not as present in volunteer movements providing aid after Spain's floods, indicate that these platforms are not exclusively focused in humanitarian and urgent needs, but that they intend to influence the future of Sudan's society. It is not so clear in the cases of Ukraine and Morocco, where crises are still ongoing or not completely resolved, added to the fact that the research depth for these two additional cases on this thesis has been less important.

[12] The slogan of 'El pueblo salva al pueblo' ('The people save the people') was popularized in social media and public discourse after the disastrous governmental response to the floods (Student volunteer, 2025).

6.4. Research Weaknesses

After interpreting the findings, it is necessary to point out constraints of this study. As previously mentioned, there is a slight information asymmetry between the two primary case studies. Furthermore, the data obtained from all interviews could be influenced by recall or desirability bias. This is why conclusions do not have a generalizability quality to them, despite bringing a relevant and innovative perspective to the humanitarian field.

In terms of the theoretical approach, the use of Kropotkin's framework from a sociological perspective provides ideological grounding, but it is empirically limited. This could result into idealized interpretations of mutual aid's impact within the results' interpretation, although intentional efforts have been dedicated to avoiding this.[13]

Lastly, there is also a risk of contextual determinism and false attributions of causality with the selected comparative design, given that the type of crisis (conflict vs. disaster) is not the only variable that could influence the impact of mutual aid. It is acknowledged that other unaccounted variables could also shape local initiatives' outcomes.

[13] The inclusion of a literature section (3.2. Theoretical Standards of Mutual Aid's Impact) identifying all the themes regarding benefits, limitations, and long-term effects of mutual aid intends to describe the analyzed phenomenon in a factual way.

7. Conclusions

In this thesis, the research question concerning the differences between the impact of mutual aid in two types of humanitarian crises – armed conflict vs. natural disaster – has been addressed through a qualitative analysis of data from Sudan's ERR and Spain's DANA volunteer groups as case studies. The main results indicate that the benefits, limitations, and long-term effects determining the impact of mutual aid responses by grassroots organizations in both the primary and secondary cases are not radically at odds, but there are some important distinctions.

We observe that in armed conflict settings, mutual aid efforts face more limitations and generate less benefits than in natural disaster contexts. Moreover, since conflicts are characterized by longer time spans where mutual aid efforts emerge while wars are still ongoing – and governmental structures remain weak –, long-term effects are more prone to materialize in these contexts. On the other hand, natural disaster emergencies last for shorter periods, with governments quickly resuming their activities and often taking over tasks that were previously carried out by mutual aid platforms (Solnit, 2009, p. 195).

It is complicated to measure the impact of mutual aid in terms of effectiveness, sustainability, and adaptability from a qualitative perspective. These factors are closely tied to context specificity, which varies greatly depending on the type of crisis setting, but also on the particular characteristics of the crisis itself (Knearem et al., 2023). Because the crises under study are ongoing/recent (2023–2025), mutual aid's long-term effects are not yet fully observable. Therefore, conclusions drawn on durability remain speculative. Overall, the findings indicate that mutual aid efforts in natural disaster settings have better chances at escalating effectiveness and sustainability, while locally led responses in armed conflict appear to be more adaptable to unpredictable environments. This illustrates why disaster and conflict settings should not be treated indistinctively.

Lastly, before introducing the recommendations, the overarching answer to the research questions is that, while mutual aid is a useful mechanism in all crisis responses, its impact does depend on the type of setting and surrounding institutional environment. In cases of natural disaster, grassroots efforts amplify benefits through rapid mobilization and relative safety, despite operational constraints. In conflict-affected contexts, mutual aid has the potential for sustained solidarity and autonomous community governance.

However, there are grave limitations due to insecurity and scarce resources. Ultimately, the long-term effects of mutual aid – ranging from resilience to political empowerment – seem to be most prone to take root in conflict zones where formal systems have collapsed.

7.1. Recommendations

To ameliorate mutual aid efforts and their impact, five recommendations stemming from the results are presented to contribute to the knowledge of humanitarian and academic fields. Since it is not possible to make definitive statements about changing/evolving scenarios, the following suggestions are presented from a speculative approach. Thus, their intention is not to modify policy decisions but rather provide insightful reflections.

- Enhance volunteer safety and protection. Considering potential measures that mitigate risks, increase protection, and ensure the physical integrity of mutual aid providers is crucial. This is especially relevant in armed conflict contexts, where dangers are more exacerbated (Madryha et al., 2024, p. 193).
- Assess long-term effects of volunteer activities on social cohesion and resilience in post-crisis contexts. Analyzing how mutual aid work impacts social integration and restoration processes after conflicts/disasters can reveal important aspects for developing recovery strategies, going beyond urgent humanitarian responses (ibid).
- Secure larger amounts of funding for local initiatives that implement mutual aid strategies. Materializing the localization agenda takes effort, and it should give agency to affected communities and grassroots organizations by providing adaptable funding. In that way, community expectations are better managed, and a commitment is established from the outset (Rose and Adler, 2024, p. 11).
- Look into the cooperation dynamics between mutual aid and formal strategies. Collaborating and building trust with government bodies and international actors should not be seen as a competing approach, but as a complementary one. Local communities have the capacity to help improve adaptability and sustainability of humanitarian assistance past emergency stages (ibid).
- Further investigate the role of technology in supporting mutual aid initiatives in crisis settings. Examining how emerging technologies, such as apps, digital platforms, and social media, can be used to coordinate volunteer efforts and improve their effectiveness (Soden and Owen, 2021, p. 15).

7.2. Final Thoughts

The framework adopted by this thesis treats mutual aid as a conceptually and operationally distinct concept, not linked to formal humanitarian responses. However, in practice, boundaries are often blurred. As an innovative approach, mutual aid's effects should be carefully studied in different humanitarian settings before encouraging it as a standardized practice or discarding its effectiveness altogether (L2GP, 2024).

To conclude, it is imperative to acknowledge the additional avenues of future research connected to mutual aid's impact in different crisis settings. These should be explored, profiting from the potential of expansion of the study of mutual aid's long-term consequences. From this thesis's results, it is resolved that mutual aid instances should be increasingly investigated as projects with a salient role within the humanitarian sector.

Bibliography

Aliño, B. (2024) 'Miles de voluntarios intentan saltarse la prohibición y seguir con las labores de limpieza', *El Mundo*, 3rd November. Available at: https://www.elmundo.es/comunidad-valenciana/2024/11/03/6727327de4d4d8d4638b457a.html.

Anckar, C. (2008) 'On the Applicability of the Most Similar Systems Design and the Most Different Systems Design in Comparative Research', *International Journal of Social Research Methodology*, 11(5), pp. 389–401. Available at: https://doi.org/10.1080/13645570701401552.

Boukhris, F. (2023) *Al-Haouz Earthquake in Morocco and Civil Society Initiatives*, *Arab NGO Network for Development*. Available at: http://annd.org/en/publications/details/al-haouz-earthquake-in-morocco-and-civil-society-initiatives-dr-fawzi-boukhris

Brun, C. and Horst, C. (2023) 'Towards a Conceptualisation of Relational Humanitarianism', *Journal of Humanitarian Affairs*, 5(1), pp. 62–72. Available at: https://doi.org/10.7227/JHA.103.

Carstensen, N., Mudhar, M. and Munksgaard, F.S. (2021) '"Let communities do their work": the role of mutual aid and self-help groups in the Covid-19 pandemic response', *Disasters*, 45(S1), pp. S146–S173. Available at: https://doi.org/10.1111/disa.12515.

Carstensen, N. and Sebit, L. (2023) 'Mutual aid in Sudan: the future of aid?', *Humanitarian Practice Network*, 11 October. Available at: https://odihpn.org/publication/mutual-aid-in-sudan-the-future-of-aid/.

Celikates, R. and Flynn, J. (2023) 'Critical Theory (Frankfurt School)', in E.N. Zalta and U. Nodelman (eds) *The Stanford Encyclopedia of Philosophy*. Winter 2023. Metaphysics Research Lab, Stanford University. Available at: https://plato.stanford.edu/archives/win2023/entries/critical-theory/.

Collado, N. and Benlloch, L. (2025) 'Más allá de la emergencia: extender el apoyo mutuo tras la Dana', *El Salto Diario*, 12 January. Available at: https://www.elsaltodiario.com/dana/alla-emergencia-extender-apoyo-mutuo-dana.

Deineko, O. (2023) 'Ukraine, War and Resistance: Reshaping Social Cohesion', *Studia Socjologiczne*, 249(2), pp. 155–177.

Dynes, R. (2006) 'Social Capital: Dealing with Community Emergencies', *Homeland Security Affairs*, 2(2), pp. 1–26.

Errazzouki, S. (2024) 'Between rubble and rage: reflections on the earthquake in Morocco and its aftermath', *The Journal of North African Studies*, 29(2), pp. 197–205. Available at: https://doi.org/10.1080/13629387.2023.2294509.

Europa Press (2024) *Voluntarios y organizaciones sociales crean una red de apoyo mutuo para afectados por la DANA*, *Actualidad Valencia*. Available at: https://actualidadvalencia.com/voluntarios-organizaciones-sociales-red-apoyo-mutuo-afectados-dana/.

Fechter, A.-M. and Schwittay, A. (2019) 'Citizen aid: grassroots interventions in development and humanitarianism', *Third World Quarterly*, 40(10), pp. 1769–1780. Available at: https://doi.org/10.1080/01436597.2019.1656062.

Glassman, M. (2000) 'Mutual Aid Theory and Human Development: Sociability as Primary', *Journal for the Theory of Social Behaviour*, 30(4), pp. 391–412. Available at: https://doi.org/10.1111/1468-5914.00137.

Hirono, M. and Nurdin, M.R. (2024) 'Local knowledge as the basis of disaster management and humanitarian assistance', *Disasters*, 48(S1), pp. 12634. Available at: https://doi.org/10.1111/disa.12634.

Kleinfeld, P. (2025) 'Eight stories about mutual aid to mark two years of war in Sudan', *The New Humanitarian*, 15 April. Available at: https://www.thenewhumanitarian.org/news/2025/04/15/eight-stories-about-mutual-aid-mark-two-years-war-sudan.

Knearem, T. *et al.* (2024) 'Solidarity not Charity! Empowering Local Communities for Disaster Relief during COVID-19 through Grassroots Support', *Computer Supported Cooperative Work (CSCW)*, 33(4), pp. 559–604. Available at: https://doi.org/10.1007/s10606-023-09484-5.

Kotani, H. and Yokomatsu, M. (2016) 'Natural disasters and dynamics of "A paradise built in hell": a social network approach', *Natural Hazards*, 84(1), pp. 309–333. Available at: https://doi.org/10.1007/s11069-016-2432-8.

Kropotkin, P. (2012) *Mutual Aid: A Factor of Evolution*. Dover Publications.

Kuka, H. (2024) 'A mutual aid volunteer reflects on a year of war in Sudan', *The New Humanitarian*, 22 April. Available at: https://www.thenewhumanitarian.org/opinion/first-person/2024/04/22/mutual-aid-volunteer-reflects-year-war-sudan.

Kynoch, A. (2025) 'Mutual Aid and Community in Times of Crisis: Building Together', *Future of Cities*, 11 January. Available at: https://focities.com/mutual-aid-and-community-in-times-of-crisis-building-together/.

L2GP (2024) *Local to Global Protection*, *Mutual Aid*. Available at: https://www.local2global.info/.

Loy, I. and Worley, W. (2025) *What's shaping aid policy in 2025?*, *The New Humanitarian*. Available at: https://www.thenewhumanitarian.org/analysis/2025/01/09/whats-shaping-aid-policy-2025.

Madryha, T. *et al.* (2024) 'The volunteer movement as a tool for strengthening civil solidarity in wartime', *Cadernos de Educação Tecnologia e Sociedade*, 14(2). Available at: https://brajets.com/brajets/article/view/1589.

Martín, J.D.L. (2024) 'Las democracias, ante las catástrofes: prevención y establecimiento de responsabilidades', *Cuadernos Electrónicos de Filosofía del Derecho*, (52), pp. 1–23. Available at: https://doi.org/10.7203/CEFD.52.29827.

Monzón, J.L. and Chaves, R. (2024) 'Editorial en homenaje a las víctimas y afectados por las inundaciones de la DANA en Valencia', *CIRIEC-España, revista de economía pública, social y cooperativa*, (112), pp. 5–9.

Naeem, M. *et al.* (2023) 'A Step-by-Step Process of Thematic Analysis to Develop a Conceptual Model in Qualitative Research', *International Journal of Qualitative Methods*, 22, pp. 1-18. Available at: https://doi.org/10.1177/16094069231205789.

Nasir, R., Rhodes, T. and Kleinfeld, P. (2023) 'In Sudan, mutual aid groups face security threats and funding gaps', The New Humanitarian, 2 August. Available at: https://www.thenewhumanitarian.org/news-feature/2023/08/02/how-mutual-aid-networks-are-powering-sudans-humanitarian-response.

Rose, J. and Adler, C.M. (2024) 'A Framework for Effective Collaboration with Crisis-Affected Communities', *Challenges*, 15(1), p. 13. Available at: https://doi.org/10.3390/challe15010013.

Saiz Saéz, J. (2024) *AyudaTerreta - Lo que ha ocurrido detrás de la plataforma*, *LinkedIn*. Available at: https://www.linkedin.com/pulse/ayudaterreta-lo-que-ha-ocurrido-detr%C3%A1s-de-la-jorge-saiz-s%C3%A1ez-tm72f/.

Sandri, E. (2018) '"Volunteer Humanitarianism": volunteers and humanitarian aid in the Jungle refugee camp of Calais', *Journal of Ethnic and Migration Studies*, 44(1), pp. 65–80. Available at: https://doi.org/10.1080/1369183X.2017.1352467.

Soden, R. and Owen, E. (2021) 'Dilemmas in Mutual Aid: Lessons for Crisis Informatics from an Emergent Community Response to the Pandemic', *Proc. ACM Hum. - Comput. Interact.*, 5(CSCW2), pp. 1-19. Available at: https://doi.org/10.1145/3479862.

Solnit, R. (2009) *A Paradise Built in Hell: The Extraordinary Communities that Arise in Disasters*. New York, United States: Viking Penguin.

Spade, D. (2020a) *Mutual Aid, Building Solidarity During this Crisis (and the Next)*. Verso.

Spade, D. (2020b) 'Solidarity Not Charity; Mutual Aid for Mobilization and Survival', *Social Text*, 38(142), pp. 131–151. Available at: https://doi.org/10.1215/01642472-7971139.

Stallings, R.A. and Quarantelli, E.L. (1985) 'Emergent Citizen Groups and Emergency Management', *Public Administration Review*, 45, pp. 93–100. Available at: https://doi.org/10.2307/3135003.

Steinberg, D.M. (2003) 'The Magic of Mutual Aid', *Social Work with Groups*, 25(1–2), pp. 31–38. Available at: https://doi.org/10.1300/J009v25n01_05.

Taithe, B. (2019) 'Demotic humanitarians: historical perspectives on the global reach of local initiatives, 1940–2017', *Third World Quarterly*, 40(10), pp. 1781–1798. Available at: https://doi.org/10.1080/01436597.2019.1630815.

Wang, J.-J. (2013) 'Post-disaster cross-nation mutual aid in natural hazards: case analysis from sociology of disaster and disaster politics perspectives', *Natural Hazards*, 66(2), pp. 413–438. Available at: https://doi.org/10.1007/s11069-012-0493-x.

Appendices

Appendix 1. *Sudan Case Study TA Table (Source: Author's own, 2025).*

Data Source	Statements / Quotations	Key Concepts	Codes	Topic (for Codes)	Themes
OCHA worker	"People who are assisting actually understand the	Understand local dynamics	3	Local participation and knowledge	Benefits
OCHA worker	"ERRs are not officially registered as CSOs, so tl	Faster reactions and no bureaucracy	4	Rapid responses	Benefits
OCHA worker	"Initial overlap with different local efforts"	Overlap	5	Lack of logistic coordination	Limitations
OCHA worker	"Expansion of mutual aid groups and strenghteni	Strengthened coordination	1	Political mobilization and collective action	Benefits
OCHA worker	"Localization has experienced an approach shift,	Bottom-up localized approach	1	Political mobilization and collective action	Benefits
OCHA worker	"Mutual aid groups are located in areas where no	Access and presence in remote areas	4	Rapid responses	Benefits
OCHA worker	"Horizontal distribution of different types of esse	Equal distribution of services	2	Building solidarity	Benefits
OCHA worker	"The insecurity and risks that these groups are fa	Insecurity and threats to life	8	Safety threats	Limitations
OCHA worker	"USAID cuts are affecting the funds that mutual	Lack of financial resources	7	Scarcity and competition	Limitations
OCHA worker	"Inevitable competition among mutual aid groups	Competition among mutual aid groups	7	Scarcity and competition	Limitations
OCHA worker	"When authorities reclaim their power, the burea	Bureaucratization of the humanitarian response	6	Legitimacy challenges	Limitations
OCHA worker	"Not all mutual aid groups are completely neutral	Neutrality and adopting political stances	1	Political mobilization and collective action	Benefits
OCHA worker	"The fact that ERRs do not support any of the gı	Protection and exposure to threats	8	Safety threats	Limitations
OCHA worker	"For the first time there is an effort to understanc	Specialized aid for the needs of the communities	3	Local participation and knowledge	Benefits
OCHA worker	"The hope for the future is that these experiences	Mutual aid as humanitarian action	12	Greater autonomy	Long-term effects
OCHA worker	"Grassroots organizations need to be involved in	Grassroots participation and adequate responses	10	Accountability and transparency	Long-term effects
OCHA worker	"The current context is not sustainable, given tha	Increasing risks and evacuation of volunteers	8	Safety threats	Limitations
OCHA worker	"Community-led initatives in Sudan cannot and h	Substitute formal actors	7	Scarcity and competition	Limitations
OCHA worker	"For ERRs, transforming into an NGO would me	Community connection and political message	1	Political mobilization and collective action	Benefits
OCHA worker	"In Khartoum, where the government is in contro	Lack of government recognition	6	Legitimacy challenges	Limitations
Sudanese journalist and volunteer	"ERRs still suffer from a certain lack of capacity	Lack of capacity and not enough donor support	7	Scarcity and competition	Limitations
Sudanese journalist and volunteer	"There have been internal difficulties to coordina	Difficulties to coordinate and lack of information	5	Lack of logistic coordination	Limitations
Sudanese journalist and volunteer	"There is an unspecified fear of corruption for dc	Fear of corruption and no permission	6	Legitimacy challenges	Limitations
Sudanese journalist and volunteer	"Safety threats are a constant concern for ERR n	Putting lives at risk and not assessing dangers	8	Safety threats	Limitations
Sudanese journalist and volunteer	"Some previous organizations and political activis	Organizational connections and building efforts	1	Political mobilization and collective action	Benefits
Sudanese journalist and volunteer	"A psychological impact can be extracted from E	Psychological support and collective care	2	Building solidarity	Benefits
Sudanese journalist and volunteer	"The first shocks of the conflict were addressed l	Local first responders and speedy reaction	4	Rapid responses	Benefits
Sudanese journalist and volunteer	"Communities are demonstrating they can be acc	Accountable communities	12	Greater autonomy	Long-term effects
Sudanese journalist and volunteer	"Transparency towards beneficiaries is achieved	Transparency towards beneficiaries	10	Accountability and transparency	Long-term effects
Sudanese journalist and volunteer	"A push for democratization and local governanc	Participative and locally-led civil society	3	Local participation and knowledge	Benefits
(Carstensen and Sebit, 2023)	"Sudanese volunteers and activists (...) will also t	Building capacities and knowledge for the future	11	Enhanced resilience and empowerment	Long-term effects
(Olson et al., 2024)	"Threats to life and well-being are common. Inte	Threats to life and insecurity	8	Safety threats	Limitations
(Nasir et al. 2023)	"Resistance committee member Mohamed Obaid	Replacement of the state in service delivery	12	Greater autonomy	Long-term effects
(ERR Khartoum State, 2025)	"For capacity building and awareness, ERR Khar	Capacity building mentorships and solidarity	9	Trust-building	Long-term effects
(Carstensen and Sebit, 2023)	"Locally led response can be much faster and mc	Fast and cost-efficient responses	4	Rapid responses	Benefits
(Kleinfeld, 2025)	"More than 50 volunteers have died during the w	Targeted killings and indirect deaths	8	Safety threats	Limitations
(Olson et al., 2024)	"Many volunteers drew hope from the ERRs' pr	Expansion of a solidarity movement	2	Building solidarity	Benefits
(Kuka, 2024)	"The Localisation Coordination Council has beco	Creation of civilian governance structures	11	Enhanced resilience and empowerment	Long-term effects
(Carstensen and Sebit, 2023)	"Coordination of spontaneous mutual aid respons	Constrained coordination	5	Lack of logistic coordination	Limitations
(Olson et al., 2024)	"ERRs have gradually built trusting relationships	Legitimacy and accountability to neighbours	10	Accountability and transparency	Long-term effects

Appendix 2. *Spain Case Study TA Table (Source: Author's own, 2025).*

Data Source	Statements / Quotations	Key Concepts	Codes	Topic (for Codes)	Themes
Mutual aid organizer	"The mutual aid network (Xarxa Suport Mutu Dana	Social movements and collective action	1	Political mobilization and collective action	Benefits
Mutual aid organizer	"Neighbours helping each other was the easiest way	Speed and access to those affected	4	Rapid responses	Benefits
Mutual aid organizer	"Lack of previous experience meant that volunteers	No experience and improvization	5	Lack of logistic coordination	Limitations
Mutual aid organizer	"There was an enormous lack of resources consideri	Lack of resources	7	Scarcity and competition	Limitations
Mutual aid organizer	"Institutions did not reply to many of the appeals the	No recognition by institutions	6	Legitimacy challenges	Limitations
Mutual aid organizer	"Brigades were organized with local volunteers and y	Local involvement and sharing of information	3	Local participation and knowledge	Benefits
Mutual aid organizer	"Localized responses by volunteers were key to prov	Faster process and localization	4	Rapid responses	Benefits
Mutual aid organizer	"It is necessary to be organized and active in our cor	Common spaces and close links with neighbours	9	Trust-building	Long-term effects
Mutual aid organizer	"Volunteer mobilization from non-affected areas to t	Volunteer mobilization and community care	2	Building solidarity	Benefits
Mutual aid organizer	"Neighborhood organizations that were already activ	Previously active organizations taking initative to provide help	1	Political mobilization and collective action	Benefits
Mutual aid organizer	"Political responsibilities have not been properly attri	Attributing political responsibility and demanding justice	10	Accountability and transparency	Long-term effects
Mutual aid organizer	"Since institutional bureaucracy does not allow a fast	Civil society's expertise to respond to crisis	11	Enhanced resilience and empowerment	Long-term effects
Spanish community member	"Most of the entries to the heavily affected neighbou	Great efforts from the volunteers to provide help	2	Building solidarity	Benefits
Spanish community member	"After a couple of days, it was clear that volunteers	Not enough capacity, need to coordinate with formal actors	5	Lack of logistic coordination	Limitations
Spanish community member	"Once the local, regional, and national professional e	Orders from authorities to stop volunteer work	6	Legitimacy challenges	Limitations
Spanish community member	"Cleaning mud and moving around the flooded areas	Mobility challenges and health hazards	8	Safety threats	Limitations
Spanish community member	"Neighbours were helping each other during and afte	Neighbour cooperation and youth mass mobilization	1	Political mobilization and collective action	Benefits
Spanish community member	"You could tell that volunteers had good intetnions a	Support and give hope to victims	2	Building solidarity	Benefits
Spanish community member	"A lot of local medical personnel and psychologists v	Locals as first responders and coordinators of aid	3	Local participation and knowledge	Benefits
Spanish community member	"At the neighbourhood level, there has been a renew	Communities uniting and distrust towards government	12	Greater autonomy	Long-term effects
Spanish community member	"In the face of governmental inaction, volunteers sho	Government inaction and help for the communities	10	Accountability and transparency	Long-term effects
Spanish community member	"Several groups of volunteers were working in affec	Months of work and reconstruction committees	11	Enhanced resilience and empowerment	Long-term effects
Student volunteer	"During the first week post-disaster, the first-respon	Local first-responders and lack of presence from formal institutions	4	Rapid responses	Benefits
Student volunteer	"People that self-organized and relied on their comm	More effectiveness, flexibility and broader reach	3	Local participation and knowledge	Benefits
Student volunteer	"Volunteers participating in locally-organized networ	Direct contact to affected families, listening to needs	2	Building solidarity	Benefits
Student volunteer	"There were access issues because to reach affected	Denied access to affected areas, government limitations	6	Legitimacy challenges	Limitations
Student volunteer	"Sometimes volunteers were sent to neighbourhoods	Poor organization and communication between networks	5	Lack of logistic coordination	Limitations
Student volunteer	"You could sense some tensions between authorities	Tensions between authorities and volunteer coalitions	7	Scarcity and competition	Limitations
Student volunteer	"Social media and 'mouth to mouth' forms of comm	Social media and easier spread of important information	1	Political mobilization and collective action	Benefits
Student volunteer	"Weeks after the floods, the sentiment of 'the people	Increased solidarity and resilience	11	Enhanced resilience and empowerment	Long-term effects
Student volunteer	"People who suffered the gravest consequences of th	Stronger networks and relying on your community	9	Trust-building	Long-term effects
Student volunteer	"Nowadays, when there is an alert for heavy rain, ev	Improved expectations for people's reactions and experience	12	Greater autonomy	Long-term effects
(Monzón and Chaves, 2024)	"During the emergency phase and the initial stages o	Resilience and capacity to respond	1	Political mobilization and collective action	Benefits
(Saiz Saéz, 2024)	"The goal of creating 'Ayuda Terreta' was clear: to h	Faster mediums to provide help for the affected people	4	Rapid responses	Benefits
(Collado and Benlloch, 2025)	"Many local organizations are supporting a reconstru	Bottom-up community reconstruction and stronger civil networks	11	Enhanced resilience and empowerment	Long-term effects
(Aliño, 2024)	"The Valencian government restricted the movemen	Restricted movement of volunteers and priority of official response	6	Legitimacy challenges	Limitations
(Europa Press, 2024)	"Solidarity and mutual support among people were a	Solidarity and progressive reach growth	2	Building solidarity	Benefits
(Monzón and Chaves, 2024)	"Even though there was no alert or danger anymore,	No danger but difficulty of accessing affected areas	8	Safety threats	Limitations
(Martín, 2023)	"The Valencian government had the necessary resou	Government had the resources, volunteers had limited capacity	7	Scarcity and competition	Limitations
(Martín, 2023)	"Affected individuals were the first ones to contribut	Affected communities acting as providers of assistance	3	Local participation and knowledge	Benefits

Appendix 3. *Standard Interview Structure (Source: Author's own, 2025).*

Section 1: General Evolution of Mutual Aid

1. How and since when have you been involved in or worked with mutual aid networks in Sudan/Spain?
2. How have mutual aid networks, evolved since the start of the crisis? What have been the main challenges in sustaining these efforts?

Section 2: Impact & Effectiveness of Mutual Aid Initiatives

3. What are the most significant benefits that mutual aid networks have provided to communities affected by the conflict/disaster?
4. How have mutual aid efforts compared to or complemented international humanitarian aid initiatives and governmental responses in Sudan/Spain?
5. What types of services and support have been most effective in strengthening community survival and resilience?

Section 3: Limitations & Risks of Mutual Aid

6. What structural or operational limitations have affected the ability of mutual aid initiatives to scale up or sustain their work?
7. Have mutual aid groups faced threats from armed actors or political forces? If so, how has this affected their ability to operate?
8. Are there risks of mutual aid initiatives being politicized, co-opted, fragmented or dismissed due to conflict dynamics or institutional pressures?

Section 4: Long-Term Impact & Sustainability

9. What long-term impact do you think mutual aid initiatives will have on Sudanese/Spanish civil society in terms of community resilience?
10. How can mutual aid initiatives in Sudan/Spain transition from emergency relief to long-term, sustainable community-based support systems?

Final Comments and Follow-up

- Do you have any additional insights on the role of mutual aid humanitarian settings?
- Would you recommend other key actors or organizations to interview for further research regarding this case study?